

Solving Equations with Squares

Name
Date
Course

Solve.

$$\begin{aligned} \textcircled{1} \quad x^2 &= 36 \\ \sqrt{x^2} &= \pm\sqrt{36} \\ \boxed{x = \pm 6} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad 8 &= \frac{y^2}{2} \\ (2)8 &= \frac{y^2}{2}(2) \\ 16 &= y^2 \\ \pm\sqrt{16} &= \sqrt{y^2} \\ \boxed{\pm 4 = y} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad 81 &= x^2 \\ \pm\sqrt{81} &= \sqrt{x^2} \\ \boxed{\pm 9 = x} \end{aligned}$$

$$\begin{aligned} \textcircled{4} \quad \frac{y^2}{5} &= 5 \\ (5)\frac{y^2}{5} &= 5(5) \\ y^2 &= 25 \\ \sqrt{y^2} &= \pm\sqrt{25} \\ \boxed{y = \pm 5} \end{aligned}$$

$$\begin{aligned} \textcircled{5} \quad 7 &= z^2 + 6 \\ -6 + 7 &= z^2 + 6 - 6 \\ 1 &= z^2 \\ \pm\sqrt{1} &= \sqrt{z^2} \\ \boxed{\pm 1 = z} \end{aligned}$$

$$\begin{aligned} \textcircled{6} \quad 9 &= 7 + z^2 \\ -7 + 9 &= 7 + z^2 - 7 \\ 2 &= z^2 \\ \pm\sqrt{2} &= \sqrt{z^2} \\ \boxed{\pm\sqrt{2} = z} \end{aligned}$$

$$\begin{aligned} \textcircled{7} \quad 4 &= 3y^2 \\ \frac{4}{3} &= \frac{3y^2}{3} \\ \frac{4}{3} &= y^2 \\ \pm\sqrt{\frac{4}{3}} &= \sqrt{y^2} \\ \boxed{\pm\frac{2\sqrt{3}}{3} = y} \end{aligned}$$

$$\begin{aligned} \textcircled{8} \quad 4x^2 &= 5 \\ \frac{4x^2}{4} &= \frac{5}{4} \\ x^2 &= \frac{5}{4} \\ \sqrt{x^2} &= \pm\sqrt{\frac{5}{4}} \\ \boxed{x = \pm\frac{\sqrt{5}}{2}} \end{aligned}$$

$$\begin{aligned} \textcircled{9} \quad 8y^2 - 7 &= 25 \\ y^2 &= 4 \\ \sqrt{y^2} &= \pm\sqrt{4} \\ \boxed{y = \pm 2} \end{aligned}$$

$$\begin{aligned} \textcircled{10} \quad -3 + 8y^2 &= 4 \\ y^2 &= \frac{7}{8} \\ \sqrt{y^2} &= \pm\sqrt{\frac{7}{8}} \\ y &= \pm\frac{\sqrt{7}}{2\sqrt{2}} \\ \boxed{y = \pm\frac{\sqrt{14}}{4}} \end{aligned}$$

$$\begin{aligned} \textcircled{11} \quad (a+3)^2 + 6 &= 8 \\ (a+3)^2 &= 2 \\ \sqrt{(a+3)^2} &= \pm\sqrt{2} \\ a+3 &= \pm\sqrt{2} \\ \boxed{a = -3 \pm \sqrt{2}} \\ \text{or } a &= \pm\sqrt{2} - 3 \end{aligned}$$

$$\begin{aligned} \textcircled{12} \quad 6 &= (a+5)^2 - 1 \\ 7 &= (a+5)^2 \\ \pm\sqrt{7} &= \sqrt{(a+5)^2} \\ \pm\sqrt{7} &= a+5 \\ \boxed{\pm\sqrt{7} - 5 = a} \end{aligned}$$

$$\begin{aligned} \textcircled{13} \quad y^2 &= 20 \\ \sqrt{y^2} &= \pm\sqrt{20} \\ \boxed{y = \pm 2\sqrt{5}} \end{aligned}$$

$$\begin{aligned} \textcircled{14} \quad 88 &= y^2 \\ \pm\sqrt{88} &= \sqrt{y^2} \\ \pm\sqrt{4 \cdot 22} &= y \\ \boxed{\pm 2\sqrt{22} = y} \end{aligned}$$

$$\begin{aligned} \textcircled{15} \quad 7 &= \frac{(x+3)^2}{9} + 2 \\ 45 &= (x+3)^2 \\ \pm\sqrt{45} &= \sqrt{(x+3)^2} \\ \pm 3\sqrt{5} &= x+3 \\ \boxed{-3 \pm 3\sqrt{5} = x} \end{aligned}$$

$$\begin{aligned} \textcircled{16} \quad -1 &= \frac{(x+6)^2}{12} - 8 \\ 84 &= (x+6)^2 \\ \pm\sqrt{84} &= \sqrt{(x+6)^2} \\ \pm 2\sqrt{21} &= x+6 \\ \boxed{\pm 2\sqrt{21} - 6 = x} \end{aligned}$$

$$\begin{aligned} \textcircled{17} \quad 2(y-5)^2 - 4 &= 2 \\ 2(y-5)^2 &= 6 \\ (y-5)^2 &= 3 \\ \sqrt{(y-5)^2} &= \pm\sqrt{3} \\ y-5 &= \pm\sqrt{3} \\ \boxed{y = 5 \pm \sqrt{3}} \end{aligned}$$

$$\begin{aligned} \textcircled{18} \quad -16 &= -3(y-5)^2 + 2 \\ 6 &= (y-5)^2 \\ \pm\sqrt{6} &= \sqrt{(y-5)^2} \\ \pm\sqrt{6} &= y-5 \\ \boxed{5 \pm \sqrt{6} = y} \end{aligned}$$

$$(29) \quad 5(y-1)^2 + 4 = 16$$

$$(y-1)^2 = \frac{12}{5}$$

$$\sqrt{(y-1)^2} = \pm \sqrt{\frac{12}{5}}$$

$$y-1 = \pm \frac{2\sqrt{3}}{\sqrt{5}}$$

$$y-1 = \pm \frac{2\sqrt{15}}{5}$$

$$\boxed{y = \pm \frac{2\sqrt{15}}{5} + 1}$$

$$(30) \quad 18(z-12)^2 - 1 = 4$$

$$(z-12)^2 = \frac{5}{18}$$

$$\sqrt{(z-12)^2} = \pm \sqrt{\frac{5}{18}}$$

$$z-12 = \pm \frac{\sqrt{5}}{3\sqrt{2}}$$

$$z-12 = \pm \frac{\sqrt{10}}{6}$$

$$\boxed{z = \pm \frac{\sqrt{10}}{6} + 12}$$

$$(31) \quad 11(y-1)^2 + 8 = 32$$

$$(y-1)^2 = \frac{24}{11}$$

$$\sqrt{(y-1)^2} = \pm \sqrt{\frac{24}{11}}$$

$$y-1 = \pm \frac{2\sqrt{6}}{\sqrt{11}}$$

$$y-1 = \pm \frac{2\sqrt{66}}{11}$$

$$\boxed{y = 1 \pm \frac{2\sqrt{66}}{11}}$$

$$(32) \quad 8 = 28(y-2)^2 + 5$$

$$\frac{3}{28} = (y-2)^2$$

$$\pm \sqrt{\frac{3}{28}} = \sqrt{(y-2)^2}$$

$$\pm \frac{\sqrt{21}}{14} = y-2$$

$$\boxed{\pm \frac{\sqrt{21}}{14} + 2 = y}$$

$$(33) \quad 24 = 27(x+2)^2 + 4$$

$$\frac{20}{27} = (x+2)^2$$

$$\pm \sqrt{\frac{20}{27}} = \sqrt{(x+2)^2}$$

$$\pm \frac{\sqrt{20}}{\sqrt{27}} = x+2$$

$$\pm \frac{2\sqrt{5}}{3\sqrt{3}} = x+2$$

$$\boxed{\pm \frac{2\sqrt{15}}{9} - 2 = x}$$

$$(35) \quad 16 = \frac{(3y+2)^2}{4} + 8$$

$$32 = (3y+2)^2$$

$$\pm \sqrt{32} = \sqrt{(3y+2)^2}$$

$$\pm 4\sqrt{2} = 3y+2$$

$$\pm 4\sqrt{2} - 2 = 3y$$

$$\boxed{\frac{\pm 4\sqrt{2} - 2}{3} = y}$$

$$(34) \quad 12 = 63(2+z)^2 + 4$$

$$\frac{8}{63} = (2+z)^2$$

$$\pm \sqrt{\frac{8}{63}} = \sqrt{(2+z)^2}$$

$$\pm \frac{2\sqrt{2}}{3\sqrt{7}} = 2+z$$

$$\pm \frac{2\sqrt{14}}{21} = 2+z$$

$$\boxed{-2 \pm \frac{2\sqrt{14}}{21} = z}$$

$$(36) \quad 17 = \frac{(1+2x)^2}{6} + 8$$

$$54 = (1+2x)^2$$

$$\pm \sqrt{54} = \sqrt{(1+2x)^2}$$

$$\pm 3\sqrt{6} = 1+2x$$

$$\pm 3\sqrt{6} - 1 = 2x$$

$$\boxed{\frac{\pm 3\sqrt{6} - 1}{2} = x}$$

$$\star (37) \quad -4 = 5\left(\frac{z}{2} - 3\right)^2 - 7$$

$$\frac{3}{5} = \left(\frac{z}{2} - 3\right)^2$$

$$\pm \sqrt{\frac{3}{5}} = \sqrt{\left(\frac{z}{2} - 3\right)^2}$$

$$\pm \frac{\sqrt{15}}{5} = \frac{z}{2} - 3$$

$$\pm \frac{\sqrt{15}}{5} + 3 = \frac{z}{2}$$

$$(2)\left(\pm \frac{\sqrt{15}}{5} + 3\right) = \frac{z}{2} \quad (2)$$

$$\boxed{\pm \frac{2\sqrt{15}}{5} + 6 = z}$$

$$\star (38) \quad \frac{(1 - \frac{x}{3})^2}{3} - 9 = 1$$

$$(1 - \frac{x}{3})^2 = 30$$

$$\sqrt{(1 - \frac{x}{3})^2} = \pm \sqrt{30}$$

$$1 - \frac{x}{3} = \pm \sqrt{30}$$

$$-\frac{x}{3} = \pm \sqrt{30} - 1$$

$$(-3) \cdot \frac{x}{-3} = (\pm \sqrt{30} - 1)(-3)$$

$$\boxed{x = \pm 3\sqrt{30} + 3}$$

$$(39) \quad -6 = 7\left(\frac{y}{4} + 2\right)^2 - 8$$

$$\pm \sqrt{\frac{2}{7}} = \sqrt{\left(\frac{y}{4} + 2\right)^2}$$

$$\pm \frac{\sqrt{14}}{7} = \frac{y}{4} + 2$$

$$\pm \frac{\sqrt{14}}{7} - 2 = \frac{y}{4}$$

$$4\left(\pm \frac{\sqrt{14}}{7} - 2\right) = \frac{y}{4} \quad (4)$$

$$\boxed{\pm \frac{4\sqrt{14}}{7} - 8 = y}$$

$$(40) \quad -6 = -14 + \frac{(3 + \frac{z}{8})^2}{5}$$

$$40 = \left(3 + \frac{z}{8}\right)^2$$

$$\pm \sqrt{40} = \sqrt{\left(3 + \frac{z}{8}\right)^2}$$

$$\pm 2\sqrt{10} = 3 + \frac{z}{8}$$

$$-3 \pm 2\sqrt{10} = \frac{z}{8}$$

$$(8)(-3 \pm 2\sqrt{10}) = \frac{z}{8} \quad (8)$$

$$\boxed{-24 \pm 16\sqrt{10} = z}$$

$$\frac{2}{1} \cdot \frac{\sqrt{15}}{5}$$